Specialized Natural Language Inference for Medical Annotations

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1. Problem Statement & Motivation

Natural Language Inference (NLI) is a subfield of Natural Language Processing focussing on the recognition of textual entailment between a premise and a hypothesis (Romanov & Shivade, 2018). This process can be framed as either a two class or three class problem. Such that given premise and hypothesis, if the hypothesis contradicts the premise, the sentence pairs are classified as contradiction, and conversely if they are in agreeance, they are classified as entailment (of hypothesis within the premise). This can then be extended to a three class problem by introducing a neutral class where the hypothesis is neither contradicted nor entailed by the premise.

In the medical domain, NLI has significant impact on medical research with applications in relationship extraction and text comprehension for biomedical texts (Seerat et al., 2024).

From electronic health records to clinical trial reports, there are many areas within the field of medicine that can benefit from precise and accurate NLI. While general NLI is well researched, domain specific NLI that require extensive knowledge remain a difficult task as it requires external context knowledge and expertise (Chowdhury et al., 2020). Natural Language Inference within the medical domain also presents challenges as data scarcity is high and distinctive language used by clinicians (especially expert annotation data) require beyond surface level text understanding (Chowdhury et al., 2020). This issue is exemplified when compared to general domain or multi-domain datasets for NLI such as SNLI or Multi-NLI.

The motivation behind this study is to evaluate and improve performance of NLI models within the domain of medical text datasets for identifying textual entailment. This includes the use of classical machine learning models such as Support Vector Machines (SVMs) and Multinomial Logistic Regression (MLR) combined with NLP methods including word embeddings, as well as transformer-based models with Bi- directional Encoder Representations from Transformers (BERT) and Distillation BERT with pre-trained domain specific tokenizers.

2. Research Hypothesis

Transformer-based models are likely to outperform traditional ML models with NLP methods given the specific context of this study and the ability utilize include pre- trained tokenizers for contextual embeddings in the medical domain.

Moreover, the bidirectional embedding nature of BERT-base models are able to capture contextual information between neighbouring tokens which can be effective in the NLI task due to the nuances within the semantic relationships of premise and hypothesis text. Furthermore, TF-IDF may not be as effective as BoW given the size of our premise text data (average length of 20 tokens).

3. Related Work & Background

The use of transfer learning for transformer based models are a common practice within NLI modeling within the biomedical domain. Namely, Seerat et al. (2024) explored and utilized seven pre-trained transformer models including bioBERT and DeBERTa variants for clinical trial records data (NLI4CT) with a classification that focused on a two class problem (entailment vs. contradiction) without the neutral class. While their highest achieving model in terms of faithfulness achieved 95%, it was low in F1-measure at around approximately 0.11. Conversely, their highest achieving F1- measure (0.61) model was low in faithfulness at approximately 11%. Using the same dataset (NLI4CT), Jullien et al. (2023) introduced a multi-hop reasoning to the same classification task and incorporated both BERT-based transformer models as well as large language models (LLM), namely

GPT-3.5. Their preliminary results showed that BioBERT outperformed GPT3.5 with an F1-score of 0.644 compared to 0.604. Furthermore, the NLI4CT dataset and our MedNLI dataset have a common characteristic traits, such that the nature of the text (annotated by clinicians) include a wide use of acronyms for disease shorthand, treatment names (Jullien et al., 2023), and other similar abbreviations.

In terms of three class predictions inclusive of the neutral class, Liang & Sonntag (2025) explored the use of chain of though LLMs for explainable NLI to verify biomedical claims (hypothesis) using scientific studies (premise). While LLMs are beyond the scope of our model for this paper due to their computational and resource cost, they are still relevant and interdependent with performances of NLI tasks.

3.1 Accomplishments

- Define Preprocess pipelines for dataset
- Remove symbols, special characters, and punctuation patterns with regex — completed
- Exploratory Data Analysis
- Apply Tokenisation
- Build random choice baseline model
- Vectorizations using TFIDF
- Vectorizations using Bag-of-Words
- Improve vectorization embeddings with Random Search
- Utilize FastTEXT word embedding library from Facebook AI — Unsuccessful
- Define & Train SVM Model
- Define and train Logistic Regression Model
- Hyper-parameter tuning for LR and SVM
- Extract Contextual Embeddings w/ transfer learned BioClinicalBert
- Define and train transfer-learned BERT model
- Define and train for DistilBERT
- Predict on Test Data
- Evaluate Results
- Train Model on BioClinicalBert — Unsuccessful due to computational limit in resource and process timeouts on accelerated environments.

4. Approach & Methodology

This study consists of building traditional NLP models and transformer models. We begin by establishing the baseline model to be the random-choice model which should theoretically have an accuracy of 33.33%. The classical NLP model used incorporate Support Vector Machine Classiers and Logistic Regression. The transformer based models uses the bi-directional encoder transformer model (BERT) to compare the differences in performance between classical NLP and machine learning approaches to transformer models.

Support Vector Machines (SVM) create maximising separation hyperplanes to make classifications. While they inherently are binary classifiers, they can be extended to the three class case. We use TFIDF in combination with the SVM model to capitalize on the SVMs ability to handle high-dimensional data.

Multinomial Logistic Regression models use a SoftMax function to output probabilistic predictions for each class. In this case, they are used with the Bag-of-Words (BoW) vectorizer.

In terms of vector embeddings, Term Frequency – Inverse Document Frequency (TF–IDF) takes the most frequently distributed terms in the document and inverses the count on the premise that less frequent occurring words are the most important. Utilizing this embedding, we can eliminate potential recursive stopwords to simplify our input. However, medical premises are very short in terms of length and tokens, so it is important to determine upper and lower bounds of frequency such that we don't down sample too much. A pitfall of this method is that some common but useful abbreviation used such as '***dx***' for diagnosis or '***sx***' for symptoms may be neglected.

Furthermore, Bag of Words embeddings (BoW) was chosen with considerations for the nature of our premise including many medical abbreviations which may be frequently repeated. If there is frequent repeating overlap for certain abbreviations, it could potentially help the model in learning

correlations between them in order to improve model performance. Conversely, A shortcoming of the BoW method is that it does not take into account the order of the words with respect to each other.

Transformer-based models were implemented with the use of transfer learning, which was a method recommended by the dataset's published author Romanov & Shivade (2018) and adopted from the methodologies of Seerat et al. (2024) and Jullien et al. (2023). We first implemented a smaller variant of the BERT model under the huggingface transformer framework called 'bert-tiny' adapted from (Bhargava, Drozd, & Rogers, 2021). The bidirectional embedding within the BERT family of models uses a deep bidirectional encoder to capture relationships in each direction around the token for context aware embeddings within the model. With this utilized a domain specific BERT-based contextual embedding tokenizer called Bio- ClinicalBERT which is pre-trained on clinical and biomedical text.

Next, we adapted a larger DistilBERT variant developed by huggingface called 'distilbert-base-uncased-finetuned-sst-2-english'. The distillation in DistilBERT refers to the way it was trained through knowledge distillation of a full BERT model. This implemented model has 67 million parameters and was trained with a learning rate of *1e-5* over 3 epochs. This configuration was recommended from the model source in huggingface.

Training for both transformer based models were conducted locally using GPU acceleration natively on M3 silicon chips with metal performance shaders (mps).

5. Dataset

The dataset selected for this study is the MedNLI dataset first proposed by Romanov & Shivade (2018) and publicly accessible via huggingface. It contains 14049 unique premise-hypothesis pairs with a training, validation, and test split of 80/10/10 percent. The premise was gathered and annotated by medical professionals and clinicians covering several medical categories such as diseases, symptom, and medication (Romanov & Shivade, 2018). The target class is evenly distributed with three outcome classes, namely: entailment, contradiction, and neutral.

Avg. Sentence Length	tokens
Premise	20.0
Hypothesis	5.8
Max. Sentence Length	**tokens**
Premise	202
Hypothesis	20

Table 5.1: MedNLI Data Summary

While this is not a large data set compared in context with large language models, data scarcity remains a present challenge in natural language inference for the medical domain (Chowdhury et al., 2020).

Herlihy & Rudinger (2021) identified that MedNLI suffers from sentence artifacts attributed from annotators ablation. This is a common issue in many medical datasets that are crowd-constructed as characteristics of the annotator within the extracted text can allow classifiers to achieve results that are statistically greater than random guess (Herlihy & Rudinger, 2021). They found and confirmed in their research that MedNLI contains such semantic artifacts within the premise sentences. This is an issue because if these semantic and lexical artifacts are present, then we can not fully attribute good model performance to the model itself learning the relationship between premise and hypothesis, but rather learning the patterns and characteristics of the specific annotators of the premise, which could lead to inaccuracies and poor generalization.

Furthermore, within these semantic artifacts, the premise of the MedNLI data has a wide variety of medical abbreviations that are used as to express a multitude of information such as symptoms, time, procedure, and health metrics, etc. that furthers the complexity of deriving meaningful relationships within the textual semantics. In the example premise given in section 5.1, it is not immediately clear to the layman that 'T3N2' refers to a tumor stage and '***dx***' stands for diagnosed.

5.1 Data Pre-Processing

Data processing for traditional NLP methods

such as Linear Regression models and Support Vector Classifiers required more extensive preprocessing than the transformer based models. For LR and SVM models, the premise and hypothesis were combined as a single text input before being vectorized with TFIDF and BoW while this step was not necessary for the transformer- based models. Premise cleaning required careful consideration for removal of special characters and symbols to retain certain useful information while removing noisy punctuation patterns. In the following example premise taken from the dataset:

Ex: "Colon CA: T3N2, dx 2 yrs ago, s/p resection ([**5-17**] LN+), mets to liver"

We must distinguish that the parenthesis and asterix between '5–17' are a formatting pattern and should be removed, however symbols like the plus sign in 'LN+' denote a positive test and should not be removed.

6. Results & Analysis

Modeling results shown in fig. 6.1 reveals that all the models outperformed the baseline model of random choice. The logistic regression model had the lowest overall weighted F1 score and Accuracy at 0.59. SVM and DistilBert showed similar performances with DistilBert being marginally higher in both F1 score (0.62) and accuracy (0.62) compared to 0.60 and 0.59. It is highlighted in table 6.1 that the compact tinyBERT model performed the best with weighted F1 score of 0.69 and accuracy of 0.69.

Model	Pre	Rec	F1	Acc
Baseline	0.33	0.33	0.33	0.33
LR	0.59	0.59	0.59	0.59
SVM	0.61	0.59	0.60	0.59
BERT*	**0.69**	**0.69**	**0.69**	**0.69**
DistilBERT	0.62	0.62	0.62	0.62

Table 6.1: Model Results

With regards to class-wise classification errors, both the SVC model and the LR model were strong in terms of correctly classifying contradictions but made more of its misclassifications in the neutral and entailment category. This could be due to the fact that the distinction between neutral and entailment within the data may not always be clearly distinguished compared to the distinction between entailment and contradiction.

Hyperparameter tuning using a random search with $k = 5$ fold cross validation found optimal hyperparameters for both SVM and LR models. The defined SVM model had a linear kernel function with a penalization term of $C = 10$. The optimal model structure found for the Logistic Regression model yielded a penalization of L2 Norm. Furthermore, for optimizing word embeddings with TF-IDF vectorization, the random search found that no TF-IDF application was most optimal. The maximum and minimum document frequency threshold found were both to be equal to one, meaning that all words were retained within the data. For the count vectorizer function for Bag-of-Words embeddings, the maximum document frequency was found to be 0.75 and minimum document frequency of 4 — meaning tokens appearing in at least 4 premises and less than 75% of the total premises will be retained. Furthermore, in both cases, we iterated through configurations of unigrams, bigrams, and tri-grams (most optimal for both). Further implementations of fastTEXT word embeddings that were pre-trained on clinical notes (BioWord2Vec) were not successfully implemented due to submodule inconsistencies between python packages within our runtime.

In terms of the tinyBERT model, the use of context-aware pre-trained tokenizers were effective in capturing the semantic relationships within the premise as it improved our intermediate modeling results. The bioClinicalBERT tokenizer extracted from huggingface was able to effectively instill contextual embeddings that are applicable to our case as it was trained on the entirety of the MIMICIII medical database. Aside from using it only as a tokenizer, we also tried to implement it on our data as a transfer learning model, however the model itself timed out on both our local GPU runtime (mps) and the cloud GPU accelerated runtime (cuda).

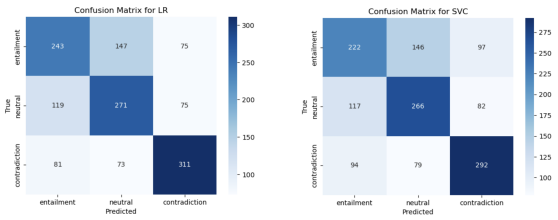

Figure 6.1: Confusion Matrices

With respected to the confusion matrices shown in figure 6.1, it is evident that the BERT model was significantly better at correctly predicting neutral and entailment compare to the SVM and LR models. While the class-wise errors for entailment and neutral are still higher compared to the contradiction class, the class-wise error rate was significantly lower from figure 6.2.

In terms of model training for transfer- learned models, trial and error for training configurations and learning rates were difficult to configure. Many negative intermediate results were produced. Following the general guideline of using a low learning rate for finetuning and a low epoch number (3 epochs) was preferred in terms of computational time and cost, as well as the risk of overfitting.

Compared to the literature surveyed for this project and the results from Seerat et al. (2024) and Jullien et al. (2023), our BERT model with a F-measure of 0.69 was able to outperformed the best BERT-based model from Seerat et al. (2023) with an F-measure of 0.61. Furthermore, it was also higher in terms of F-score compared to the preliminary baseline results of their LLM GPT3.5 model (0.604) and the bioBERT model at 0.644 from Jullien et al. (2023).

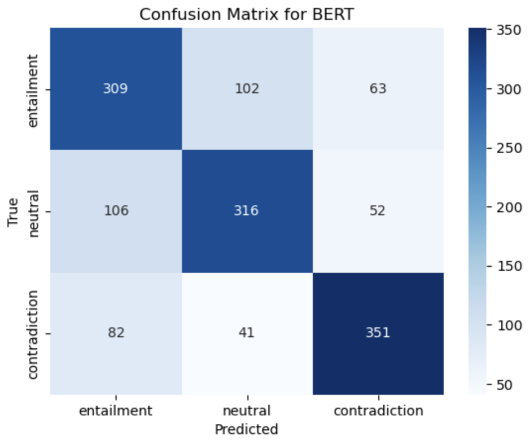

Figure 6.2: Confusion Matrix for BERT

Although the BERT model faired comparably with similar research outcomes, the model is still not achieving high enough results.

Furthermore, the NLP trials and approaches in this study do not mitigate domain specific NLI issues in the medical field well with high performance. In light of this, the NLP approaches and models experimented in this study do not effectively mitigate issues within the task of medical NLI. However it does serve as a basis for moving towards the use of LLMs within the medical field.

7. Lessons Learned & Conclusions

As we hypothesized in section 2, the BERT models were indeed more performant compared to the machine learning models. Furthermore, the use of pre-trained contextual embeddings were able to improve results in terms of F1 score and accuracy measures. Furthermore, it was surprising that the gridsearch determined no TF-IDF reduction was the most optimal — while it was expected that TF-IDF may not perform well on the premise data as each premise only had an average length of 20 tokens, it was an interesting finding to see that the randomized grid search pointed to keeping the entire feature space.

Further modeling on similar medical NLI datasets should be conducted to measure generality across different datasets.

The use of FastTEXT as a pre-trained word embedding would have been an insightful experimentation in terms of comparing with BoW and TF-IDF. Furthermore, the comparison of more BERT based models such as BioBERT could introduce more variety in terms of comparison within the BERT family. Moreover, small open sourced LLMs would be a natural extension to the next step of modeling if computing resources allowed for the training of these models.

Given that the trajectory of development in utilization of medical focused large language models from Pre-trained Language Models (Nazi & Peng, 2024), the basis of in NLI inference should present a good basis for progressing towards furthering clinical language tasks.

As Monea & Marginean (2021) stated, MedNLI is currently the largest dataset for natural language inference within the clinical domain. With the given data limitations mentioned previously, an NLP framework or pipeline for effectively defining NLI data in the medical domain for both training and benchmarking would be a useful contribution towards medical NLI as a whole.

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